Ass-1

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Q1

a) $R_t = \frac{P_t - P_{t-1}}{P_{t-1}}$

Given prices

100, 108, 103, 115, 110

month 1

$R_1 = \frac{108 - 100}{100} = 0.08 = 8\%$

month 2

$R_2 = \frac{-5}{108} = -4.63\%$

month 3

$R_3 = 11.65\%$

month 4

$R_4 = -4.35\%$

b) log returns

$r_t = \ln\left(\frac{P_t}{P_{t-1}}\right)$

month 1

$r_1 = \ln\left(\frac{108}{100}\right) = \ln(1.08) = 7.70\%$

month 2

$r_2 = \ln\left(\frac{103}{108}\right) = -4.74\%$

month 3

$r_3 = \ln\left(\frac{115}{103}\right) = 11.02\%$

month 4

$r_4 = \ln\left(\frac{110}{115}\right) = -4.45\%$

c)

month	Simple Return	Log return	Difference
1	0.08	0.07	0.003
2	-0.0463	-0.0474	0.0011
3	0.1165	0.1102	0.0063
4	-0.0435	-0.0445	0.001

The approx is least accurate in month 3 because return mag is largest

Q2

$E[R_1] = 15\%$, $E[R_2] = 7\%$

$w_1 = 0.4$, $w_2 = 0.6$

a) linear operator

$E(ax+by) = aE[X] + bE[Y]$

$E[RP] = E[W_1R_1 + W_2R_2] = W_1E[R_1] + W_2E[R_2]$

$E[RP] = [0.4 \times 0.15] + [0.6 \times 0.07]$

$= 0.102 \approx 10.20\%$

b) No, this result does not change

Q3.

$RP = W_1R_1 + W_2R_2$

let $X = W_1R_1$

$Var[X+Y] = Var(X) + var(Y) + 2COV(X,Y);$

$\sigma_P^2 = var(W_1R_1 + W_2R_2) = var[W_1R_1] + var[W_2R_2] + 2COV(W_1R_1, W_2R_2)$

$Var(ax) = a^2 \cdot var(x)$ $\quad cov(ax, by) = ab\ cov(R_1R_1)$

$\sigma_P^2 = W_1^2 var(R_1) + W_2^2 Var(R_2) + 2W_1W_2\ cov(R_1R_2)$

$var(R_1) = \sigma_1^2 \quad var(R_2) = \sigma_2^2$

$\rho_{12} = \frac{cov(R_1R_2)}{\sigma_1\sigma_2}$ $\quad COV(R_1R_2) = \rho_{12}\sigma_1\sigma_2$

$\sigma_P^2 = W_1^2\sigma_1^2 + W_2^2\sigma_2^2 + 2W_1W_2\rho_{12}\sigma_1\sigma_2$

Q4

$\mu_A = 18\%$ $\quad \sigma_A = 25\%$ $\quad \mu_B = 8\%$ $\quad \sigma_B = 12\%$

a) σ_P for $\rho = +1, -1, 0$

For $\rho = +1$

$\sigma P^2 = (0.5)^2(0.25)^2 + (0.5)^2(0.12)^2 + 2(0.5)(0.5)(0.25)(0.12)$

$\sigma_P^2 = 0.034225$

$\sigma_P = \sqrt{0.034225}$

For $\rho = 0$

$\sigma_P^2 = 0.015625 + 0.0036 + 0 = 0.013225$

$\sigma_P = \sqrt{0.015225} \approx 0.1$

For $\rho = -1$

$\sigma_P^2 = 0.015625 + 0.0036 - 0.0150 = 0.004225$

For $\rho = -1$

$\sigma_p^2 = 0.015625 + 0.0036 - 0.0150 = 0.004225$

$\sigma_p = \sqrt{0.004225}$

b) when $\rho = +1$

Proof :-

$\sigma_p^2 = w_A^2 \sigma_A^2 + w_B^2 \sigma_B^2 + 2 w_A w_B (1) \sigma_A \sigma_B$

$\sigma_p^2 = (w_A \sigma_A + w_B \sigma_B)^2$

$\boxed{\sigma_p = w_A \sigma_A + w_A \sigma_B}$

c) $\rho = -1$

$\sigma_p = |w_A \sigma_A - w_B \sigma_B|$

To absolute zero ($\sigma_p = 0$) enforce $w_A \sigma_A - w_B \sigma_B = 0$

$w_B = 1 - w_A$:

$w_A \sigma_A - (1 - w_A)\sigma_B = 0$

$w_A (\sigma_A + \sigma_B) = \sigma_B$

$w_A = \dfrac{\sigma_B}{\sigma_A + \sigma_B} = \dfrac{0.12}{0.25 + 0.12} = \dfrac{0.12}{0.37} = 0.32$ $= 32.43\%$

Q5 $\sigma_p^2 = w_1^2 \sigma_1^2 + w_2^2 \sigma_2^2 + 2 w_1 w_2 \rho \sigma_1 \sigma_2$ when ($\rho < 1$)

if the asset were positively completed ($\rho = 1$), the voilence would expand and to a perfect square, resolving exactly to the weighte average risk: $\sigma_{p\,max}: w_1 \sigma_1 + w_2 \sigma_2$

$\sigma_p^2 < w_1^2 \sigma_1^2 + w_2^2 \sigma_2^2 + 2 w_1 w_2 \sigma_1 \sigma_2 = (w_1 \sigma_1 + w_2 \sigma_2)^2$

$[\sigma_p < w_1 \sigma_1 + w_2 \sigma_2)$

Q6 $\sigma_1 = 0.25$ $\sigma_2 = 0.12$ $\sigma_3 = 0.04$

$\rho_{12} = 0.4$ $\rho_{13} = 0.1$ $\rho_{2.3} = 0.2$ $\rho_{ii} = 1.0$

$$\Sigma_{ij} = \rho_{ij}\, \sigma_i \sigma_j$$

$$\Sigma_{11} = 1.0 \times 0.25 \times 0.25 = 0.625$$

$$\Sigma_{22} = 0.0144$$

$$\Sigma_{33} = 0.0016$$

$$\begin{bmatrix} 0.0625 & 0.0120 & 0.0010 \\ 0.0120 & 0.0144 & 0.0096 \\ 0.0010 & 0.00026 & 0.0016 \end{bmatrix}$$

$$\Sigma_{12} = \Sigma_{21} = 0.0120$$

$$\Sigma_{13} = \Sigma_{31} = 0.0010$$

$$\Sigma_{23} = \Sigma_{33} = 0.00096$$

b) $\Sigma = \Sigma^T \qquad \Sigma_{ij} = \Sigma_{ji}$

So, yes it's symmetry

Q7

a)
$$SR = \frac{E[R_p] - R_f}{\sigma_p}$$

For X: $SR_x = \dfrac{0.14 - 0.04}{0.22} = \dfrac{0.10}{0.22} \approx 0.4545$

$$SR_y = \frac{0.09 - 0.04}{0.10} = \frac{0.05}{0.10} \approx 0.500$$

$$SR_z = \frac{0.20 - 0.04}{0.35} = \frac{0.11}{0.35} = 0.4571$$

b) $y > z > x$ (Potfolio Ranking & Preference)

They prefer y because it gives height return rate with risk of (0.50)

$$E(R\ given) = R_f + \sigma\ target \times SR_y$$

$$= 4\% + (35\% \times 0.50)$$

$= 21.5\%$

Since 21.5% exceeds potfolio Z's 20%

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a) Gmv potfolio is the unique allocation on the minimum variance frontin that achieve the absolute lowest possible variance among all potfolio built from risky asset

b) The students's argument is incorrect:-
The lower half of the minimum variance is strictly inficient (or any portfolio on the bottom curve, there is a corresponding Potfolio on the top half that store the exact same level of risk but offers as lighter expected return

c) CML

$$\text{slope of cml} = \frac{E(R_m) - R_f}{\sigma_m}$$

This represted the shape Ratio of the market Potfolio which serve as the market price of risk.

9.

a) $N = 36$ $\quad \sigma_i = 30\% = 0.30$

$$SE(\hat{\mu}_i) = \frac{\sigma_i}{\sqrt{N}} = \frac{0.30}{\sqrt{36}} = \frac{0.30}{6} = 0.05 = 5\%$$

b) Man variance optimization act as "error maximizer" The optiminum target assert that look highly appealing ...

"error maximiser" The optimizer target assert that look highly appealing due to overestimated historical return or understand risk

c) $N = 50$ asset

Total parameter $= \frac{N(N+1)}{2} = \frac{50 \times 51}{2} = 1275$ parameter

c) Estimating 1275 parameter accurately requires an immense amount of historical data